4.3.1 Series and parallel circuits

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	Kirchhoff's second law; the conservation of energy	
(b)	Kirchhoff's first and second laws applied to electrical circuits	
(c)	total resistance of two or more resistors in series; $R = R_1 + R_2 + \ldots$	
(d)	total resistance of two or more resistors in parallel; $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots.$	
(e)	analysis of circuits with components, including both series and parallel	
(f)	analysis of circuits with more than one source of e.m.f.	